# Machine Learning

## Laboratory on scikit-learn

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# Set-up on Google Colab

If you prefer to run the notebook in the cloud, use the `Google Colab` link provided in the repository or scan the QR code below.



Figure 1: QR code to the repository

No local installation is required; everything runs in your browser.

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# Preliminaries (on your PC)

**1. Download the Material**

Download or clone the course repository from

https://github.com/steveazzolin/machine-learning-lab26-27

and download `main.ipynb` from Google Drive.

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# Preliminaries (on your PC)

**2. Install Jupyter Notebook**

Open your terminal and install Jupyter using `pip`:

```
pip install jupyter
```

*Note:* If you are using Anaconda, Jupyter is already included.

# Preliminaries (on your PC)

**3. Navigate to the Project Folder**

Use the terminal to move into the `sklearn` directory:

```
cd path/to/sklearn
```

Then launch Jupyter Notebook:

```
jupyter-notebook
```

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## 4. Open Jupyter in Your Browser

Once Jupyter starts, your browser will open automatically showing the current working directory.

Locate and open the notebook file:

**`main.ipynb`**




Figure 2: Jupyter file browser

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# Running Cells in Jupyter

To execute a cell, you can either:

- Click the **Run** button in the toolbar, or
- Press **Shift** + **Enter** on your keyboard.

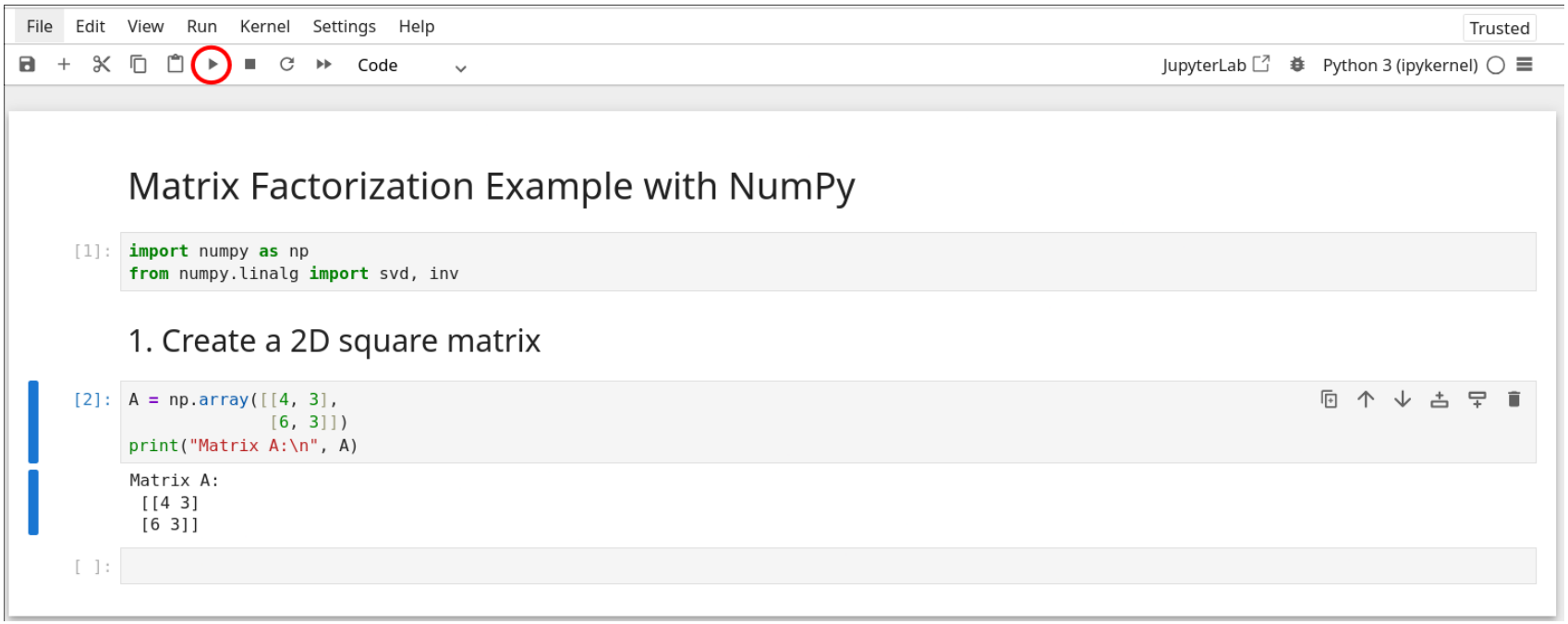



Figure 3: Running a cell in Jupyter

# Take-home Exercise

As a self-assessment exercise, complete a classification task using **Scikit-Learn**.

**Steps:**

1. Choose a dataset.
2. Train and tune a classifier (e.g., perform hyperparameter optimization).
3. Evaluate the model on the test set.
4. Compare performance across different classifiers.

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# Suggested Datasets

You can find datasets in the UCI Machine Learning Repository:

- Spambase
- Optical Recognition of Handwritten Digits
- Abalone
- Additional datasets available at https://archive.ics.uci.edu/

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# Model Comparison

Compare the performance of multiple classifiers such as:

- Decision Tree
- Random Forest
- SVM

Perform hyperparameter tuning and report the evaluation metrics (e.g., accuracy, precision, recall, F1-score).

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